

Faculty of Computing



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[Computer Communications & Network]

Lab No 6 Tasks

Task 1:

Write the IP address 222.1.1.20 mask 255.255.255.192 in CIDR.

1. Convert subnet mask to binary:

255.255.255.192 = 11111111.11111111.11111111.11000000

Here the no of one's are 26.

2. Count the number of 1's:

26 bits

So, CIDR = /26 → **222.1.1.20/26**

Task 2:

Write is the IP address 135.1.1.25 mask 255.255. 248.0 in CIDR notation.

1. Convert subnet mask:

255.255.248.0 = 11111111.11111111.11111000.00000000

2. Count 1's → 21 bits

CIDR = **135.1.1.25/21**

Task 3:

You have been allocated a class C network address of 201.1.1.0 how may hosts can you have?

Class C default subnet mask = 255.255.255.0 → /24

Formula for hosts:

$2^{(32 - \text{subnet bits})} - 2$

$2^{(32-24)} - 2 = 2^8 - 2 = 256 - 2 = 254$

254 Hosts

Task 4:

You have been allocated a class A network address of 21.0.0.0. You need create at least 10 networks and each network will support a maximum of 100 hosts. Would the following two subnet masks Work.

255.255.0.0 and or 255.255.255.0

Option 1: 255.255.255.0 (/24)

Hosts per subnet: $2^8 - 2 = 254$, Enough

Subnets available:

Class A default is /8, /24 uses 16 more bits for subnetting

$2^{16} = 65536$ subnets ,More than 10

Works

Option 2: 255.255.0.0 (/16)

Hosts per subnet: $2^{16} - 2 = 65534$ ✓□ (too many)

Subnets: $2^8 = 256$ ✓□ Enough for 10

Wastes too many IPs per subnet

Technically works but not efficient

Task 5:

You have been allocated a Class B network address of 129.1.0.0. You have subnetted it using the subnet mask 255.255.255.0 How many networks can you Have and how many hosts can you place on each network?

Default for Class B = /16

Subnet mask /24 means 8 bits used for subnetting

$2^8 = 256$ subnets

Hosts per subnet = $2^8 - 2 = 254$

256 subnets, 254 hosts per subnet

The End